

Why Parseval Does Not Evaluate a Prescribed Phase Atom

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Abstract

Continuous trigonometric polynomials can have bounded L2 norm and arbitrarily large value at a prescribed point. The normalized Dirichlet kernels give an explicit witness. Thus Parseval plus Chebyshev cannot prove the named-atom direct-twist target without additional arithmetic structure. The classification is OBSTRUCTION_L1; no L2 arithmetic progress or endpoint credit is claimed.

1 Frozen target

The selected route is `0161.direct_additive_twist_fixed_atom` [1]. The required signature is actual fixed- h_0 packet, named fixed atom, deterministic every prefix, deterministic every scale, fixed- X power at that atom, and actual active support. Throughout $h_0 = 2$ is only a source-backed data fact.

2 Exact result

Theorem 1. *For $D_m(\alpha) = \sum_{j=0}^{m-1} e(j\alpha)$, $\|D_m\|_2 = \sqrt{m}$ but $D_m(0) = m$. Point evaluation is therefore not L2-bounded, even within trigonometric polynomials.*

Proof. Orthogonality gives the squared L2 norm of D_m as m . At $\alpha=0$ all m summands equal one, so $D_m(0) = m$. If point evaluation were L2-bounded by a universal C , then $m \leq C \sqrt{m}$ for every m , a contradiction. $\square$

The smallest missing literal theorem is POINTWISE_ARITHMETIC_INPUT_BEYOND_PHASE_L2.

3 Scoped stop and claim firewall

The cell `PARSEVAL_CHEBYSHEV_TO_PRESCRIBED_ATOM` is STOP-SCOPED. This does not stop either O161 pointwise parent or the global architecture.

The result is L0/L1 only. Phase averages and Lebesgue-almost-every phase are not named atoms. Fixed $h_0 = 2$ is not decay. Archive addressing is not canonical physical representation, and empty-domain truth is not actual active support. The named-atom endpoint charge is zero, so the strict $1/400$ budget is unpaid. We claim no program-positive L2, prime-pair lower bound, or twin-prime theorem.

4 Reproducibility

The adjacent JSON payload records the full six-axis signature, source lock, typed result, exact scoped-stop cell, endpoint ledger, and claim firewall. The audit artifact contains eight true mutation-rejection assertions.

References

- [1] Liang Wang. Mvp8 source-phase route decision, 2026.